CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. Examination (CE/IT)

November 2013

IT205 OPERATING SYSTEM

Date: 14.11.2013, Thursday **Time: 01:30 p.m. To 04:30 p.m.** **Maximum Marks: 70**

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 (a) State whether the following statements are True or False. **[07]**

1. Deadlock means non availability of the primary memory.
2. It is not feasible to actually implement the optimal page replacement algorithm.
3. A process can move from blocked state to running state.
4. Each process has an own address space and single thread of control.
5. One thread can read, write or wipe out another thread's stack.
6. External Fragmentation means total memory space exists to satisfy a request, but it is not contiguous.
7. When required page is not found in secondary memory leads to page fault.

Q - 2 (a) A disk has 1000 cylinders, numbered 0 to 999. The drive is currently serving a request at cylinder 135, and the queue of pending requests is: **[05]**

89, 470, 913, 775, 948, 509, 33, 11, 134, 230

Starting from the current head position, what is the total distance (in cylinders) that the disk arm moves to satisfy all the pending requests for Elevator and C-Look disk-scheduling algorithms? Previous request was at cylinder 100.

(b) Elaborate the concept of race condition. Explain the terms: mutual exclusion, critical region. **[05]**

(c) What is deadlock? Which are the conditions for resource dealocks? **[04]**

OR

(b) Explain secret-key and public-key cryptography. **[05]**

(c) What is page table? What is the significance of it? Mention fields of Page Table Structure. **[04]**

Q - 3 (a) Discuss Peterson's Solution to achieve mutual exclusion. **[05]**

(b) Explain Redundant Array of Independent Array concept for parallel I/O operations. **[05]**

(c) Explain file system layout stored on the disk. **[04]**

OR

Q - 3 (a) Explain the Banker's algorithm for single resource to avoid deadlock. **[05]**

(b) Draw and explain 5-state process transition diagram. **[05]**

(c) Explain the three types of file structure. **[04]**

Q - 4	(a)	**Select the most appropriate option:**	[05]

1. Elevator algorithm is also known as
   (A) Optimal page replacement algorithm (B) WS Clock algorithm
   (C) Shortest Seek Time First algorithm (D) SCAN algorithm
2. Sector 0 of the disk is called
   (A) i-node (B) MBR
   (C) FAT (D) UDF
3. The amount of CPU time a process has actually used since it started is called its
   (A) total burst time (B) total execution time
   (C) total waiting time (D) current virtual time
4. The virtual address space is divided into fixed-size units are called
   (A) Page frames (B) MMU
   (C) Pages (D) Page Fault
5. Threads are also known as
   (A) Interrupts (B) Device controllers
   (C) mini-processes (D) Segmentations

	(b)	Average process size is $s = 4$KB. Each page entry requires $e = 8$ bytes. Calculate (a) optimum page size p (b) approximate no of pages needed per process.	[02]
Q - 5	(a)	Explain the role of an operating system as a resource manager.	[05]
	(b)	Explain the operation of Direct Memory Access transfer.	[05]
	(c)	Write a shell script that read an integer from the user and finds factorial.	[04]

OR

Q - 5	(a)	Explain any two process scheduling algorithms using three processes.	[05]
	(b)	Differentiate: Starvation vs. Deadlock.	[05]
	(c)	Explain the commands: ls, cat, chmod, grep	[04]
Q - 6	(a)	With reference to FIFO page replacement algorithm, explain Belady's anomaly for available page frames 3 and 4 using example.	[07]
	(b)	Discuss the Dining Philosophers problem with reference to processes.	[07]

OR

Q - 6	(a)	Explain Producer-consumer problem and its solution using semaphores.	[07]
	(b)	Explain monolithic and microkernel system structures.	[07]

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